Part I-A

1. (C) 3

2. (C) Femur

3. (A) Closing of Sutures

4. (A) Male Skull

5. (A) Identify individual.

6. (C) Head Shape.

7. (D) Facial Recognition (1)

8. (B) Affects the overall appearance of the face. (1)

9. (A) Age at death. (1)

10. (A) They are unique to each individual (1)

Part I-C.

8

20.1. Odontology :- Study of teeth is called odontology.

Role in Mass disaster situations

Odontology is very reliable evidence in forensic investigation because it has characteristics like :-

- No two individual have same teeth size.
- Morphology is vary person to person
- Size & Shape also vary person to person.
- We can extract DNA from the teeth so that we find the personal identity of a person.
- Everyone have unique pattern

In Mass disaster situation forensic experts follow steps are as follow.

- they compare 1000 to 100 of Post Mortem & Anti-Mortem data of the teeth
- By comparing find out the personal identity of a person.
- In mass disaster we found many part of to to skeletal of different persons.

Dental Records are very helpful to find out the accurate victim Identification.

We have 3 steps include in it!.

1. Post Mortem
2. Anti Mordem
3. Dental ~~Comparison~~ ~~or~~ Comparing Identification.

Post Mortem !-, 1. Take out the body ~~from~~ ;
2. Then take photographs & videos.
3. Then Record all the data about the teeth.
4. Then extract the ~~Dental~~ teeth from the skull.
Then further ~~Examine~~ the dental Records.

Anti-Mortem ;. It is the most difficult steps.
- It is done in very short time of period.
- It is not accurate properly.
- By the help of Relative, family member collect all the data about the individual.

Dental comparing identification :- Compare both Anti-Mortem and post-mortem data.

we can extract DNA from teeth by which we can find the Remaining parts belong to which ~~part~~ family.
There are 2 method of extracting DN.

1. Cryonial method :- In the teeth is cool down in low temperate.
Then grind it and Made the powder of this part.

2. Grinding method :- In this teeth are grinding the pulpal part of the teeth is grind by micromotor on forensic needle.

Facial Reconstruction

Facial Reconstruction technique is used to reconstruct the face of a person to find out their personal identity.

There are various facial ~~superimposition~~ techniques are there.
Superimposition

1. Photography superimposition :- In this superimposition we have to take a photograph of a person before death to match the details of the present & post Mortem & ante Mortem data.

Strength :- It is easy & and give accurate result & .
It help to find out the personal identity of a person.
It help in mass disaster cases.

Limitation :- It is very time consuming.
It is very difficult to collect ante Mortem photograph of a person.

2. Vediography superimposition :- In this we take a live vedio or a vedio recording of person before death by which we can compare the facial structure or features of a person.

Strength :- It help to find out the identity of person that in video & in the after Post Mortem. If details are same the person's personal identity will be known.

limitation :- It has very high chances of mistake
It did not give 100% accurate results.

3. Radiography superimposition :- In this we compare the X-Rays of a skull.

Strength :- By this we can get the ante mortem and other details of the face.
It help to recognize the person from the skull.

limitation :- It is very time consuming
- It do not make accurate face because we did not know the shape and the features like tissue depth so overall feature is not known.
- It is very difficult to find out the X-Rays of a unknown person.

Their use in both confirmed and disputed identity cases that if we have a confirmed case we also have the ante mortem data of a person so it become easy to

Reconstruct the face easily but is some cases like disputed identity cases we have to make a facial reconstruction By own measurment I ideas it does not accurate result but we get the idea that the skull belongs to that person.

Controller of Examination

18.1. Somatometric measurement. (7.5)

In this we have to measure the length breath of a bones of the Skull Skeletal or living persons.

we have many instrument to measure

1. Anthropometer.
2. sliding caliper etc.

we have different formula for each part of the face. like:

1. head

For the measurement we meet cephalic index.

Formula : $\frac{\text{head length}}{\text{head breadth}} \times 100$

2. Nose

For Nose we have nasal Index.

Formula → $\frac{\text{Nose height}}{\text{Nose breadth}} \times 100$

3. Face

For Face we have facial index.

Formula → $\frac{\text{Face breadth}}{\text{Face height}} \times 100$

4. Ears

For ears we measure height of the ear and the shape & note that the ear lobe is attached or not.

By these formulas and by these methods we can get the accurate result and made a facial feature of person to find out the personal identity

It is reliable but not 100%. because the the estimations we could not make accurate face of a person but we can get the idea of the identity of the person.

In forensic examination it has very important in the cases like mass disaster or burned cases we did not have the body of the person So by this we can identify the person's identity

17.1 Skull.

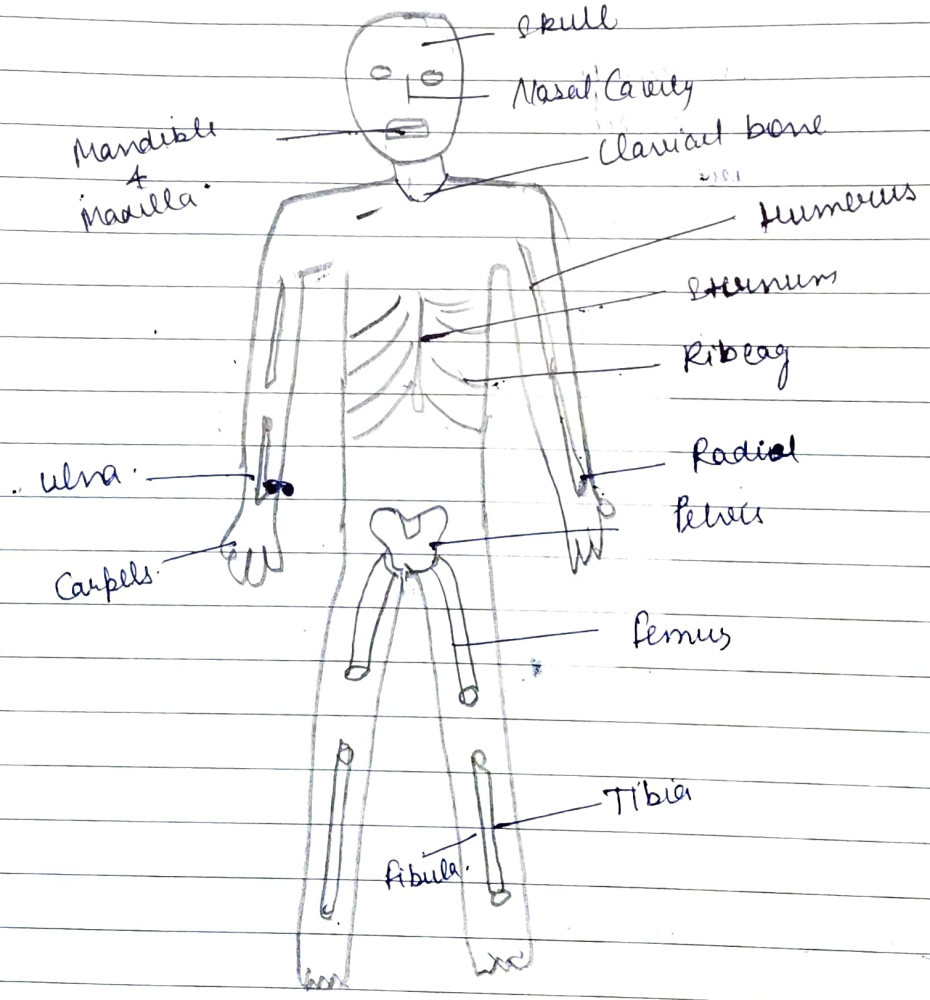
Skull is a part of the skeletal which provide shape to the face of a person.
It also protect the brain
It is very hard in nature.
By the help of skull we can determine the sex of a person.

Features	Male	Female.
General Appearence.	large	small.

. Architecture	Riged.	Rough.
Supra orbital	Highly pronounced.	less pronounced.
Orbit	Round with	Rectangle with sharp
	Sharp margin.	margin.
Gonial angle	Acute	Obtuse
Cheek. shape	Round	oval.
zygomatic bone	more prominent	less prominent.
head.	bigger	smaller.
Robusticity	Highly pronounced.	less pronounced.
Nasal cavity	sharp & pointed	triangular
Frontal eminities	Small	large.
supra orbital	thin & smooth	thick & Smooth
Ramaus height	large.	small.
Ramus structure	flattened	Rough.
Size.	bigger.	smaller.

By these characteristics we can differential the difference b/w the Male & the female skull. If we get characteristics similar to female then it is female or vice versa.

16.2


Skull
Nasal Cavity
Mandible & Maxilla
Clavicle bone
Humerus
Sternum
Ribcage
Radial
Ulna
Pelvis
Carpels
Femur
Tibia
Fibula

Skeletal.

It is structure of the body
It give shape to the body.
It is hard in nature.

⇒ We have divided the Skeletal ~~bones~~ in ~~2 part~~ parts.

- Axial
- Aperdicular

⇒ We have ~~a~~ 6 type of bones.

1. Small bone
2. Large bone
3. Pneumatic bone
4. Sesamoid bone
5. Flat bone
6. Irregular bone

⇒ We have 2 large bone in skeletal

1. Femur
2. Humerus

⇒ We have short bones are
- carpel
- tarsel

→ We also have different type of sutures present in the skull which help to get the estimation of age of a person.
→ By the help of femur bone we can determine the age of a person.
→ The formation of Bone is called ossification. It has two types ① Intramembranous ossification
② endochondral ossification
→ The cell which help in bone formation are Osteblasts. It is mainly found in disaster & burned cases.
→ By the help of skeletal parts like skull, pelvis Mandible, we can determine the sex of a person.

Part I-B.

15.2 Facial Reconstruction

Facial Reconstruction is a process in which we can find

Out the personal identity of person by reconstruct thier face from skull.

⇒ Facial Reconstruction is of 3 types.

1. 2D Facial Reconstruction
2. 3D Facial Reconstruction
3. Superimposition

⇒ 2D Facial Reconstruction !. In the we construct the face by the clay models on by ~~seeing~~ the ~~pro~~ sketching. ~~On~~ Factors.

⇒ 3D Facial Reconstruction !. In this we construct the image by computers.

⇒ Superimposition !. In this we impose the potograph & match the facial features of the person.

⇒ Factors that affect the facial Reconstruction

- Time gap → The time gap become more so we could not remember the face.

- Ageing :- It create lot of affect in ~~creating~~ constructing a image
- Memory loss
- Traumatized
- Change in facial features.

These are feature which make difficult to construct a face.

⇒ In this we have both digital & manual methods

Digital	Manual
• It is helpful in nowadays a lot of AI tools or software are launched for facial reconstruction.	• It is done in past years where we did not have time
• It ~~is not~~ give accurate result	• It take lot of time to make faces.
• It is more reliable	• It did not give accurate results
• It is less time consuming.	• It is not 100% reliable.

14.) Portrait Parle / Bertillon System.

In this system we can record the details of the criminals for further investigation.

In this system we can take the photographs in different view 1. Front view (close-up image)
2. Right side view
3. left side view

It is invented by Alphonse bertillion in 1950

In this the criminal write their detail on the board and after that we take photograph with board of them.

After this a Report is made in which all the details are mentioned with photographs.

Biometric methods.

In this system we record fingerprints of a criminal for further examination.

In this we take fingerprints and eye scan to identify the personal identification because fingerprints are never same of two person

It is a individual characteristics.

Diff.

Bertillon.

- It is used in past when the technology is not develop properly.
- It does not reliable because any person can change them.
- Face does not have same life due to some issue it may be changed.
- It is ~~her~~ time consuming.
- It is not reliable in court because any one can edit the images

Biometric

- It is new technology which is used nowdays.
- It is more reliable because every person have ~~their~~ unique points
- It does not change by the passing of time it remain same till death.
- It is very less time consuming.
- It is reliable in court.
- Access to this is not easy.

13.2 Somatometry.

It is a technique to take the measurement of the bone of the skull

By this we can Reconstruct the face of a person for identification

⇒ we have many tools to take measurement.

- Anthropometer
- Flexible tape
- Sliding Caliper
- Vernier caliper
- measuring tape
- Scale

⇒ we have different formula for different part of the bone like for head we need cephalic index.

- Nose → Nasal index
- Face → facial index

and we also include various features like ear or the other part to make the facial

Forensic Significance.

By this we can find out the measurent of bone & we can get the accurate face of a person.

In cases like mass disaste and burned case if the identity is unknown of a person we can get to know the personal identification of a person.

Formulas.

Cephalic Index → $\frac{\text{Head breadth}}{\text{Head length}} \times 100$.

Nasal Index → $\frac{\text{Nose height}}{\text{Nose breath}} \times 100$

Facial Index → $\frac{\text{Face breadth}}{\text{Face height}} \times 100$

By these index we can get the measurement.

12.1 Skull.

It is a part of skeletal.
It give shape to the face.
By the help of skull we can determine the sex of a individual.

Characteristic feature of Male skull.

Feature	Characteristics.
General appearence	large in size.
Zygomatic bone	more prominent.
supra orbitals	bigger & highly pronounced
Gonial angle	Acute
Nasal cavity	Sharp & pointed
Size	large
Head Size	bigger.
Architecture	Rigged.
Robusticity.	highly pronounced.
Ramus structure	flattened.

1102. Anthropology.

It is a study of bones

The father of Anthropology is ~~Aphon~~ Alphonse Bertillon

Forensic Significance.

- In forensic science Anthropology plays very important Role like in the cases where we can only get the skeletal It is become very difficult to solve the case or get the identity of a person or there are some case that happen where there is no dead body or any thing related to victim so then we have to use the skeletal or bones of the person for examination.

By the Anthropology we can get many informations like

By dental we can get DNA.

By skull we can get to know the age estimation & sex determination

By the skull we also Reconstruct the face of a person it help to find out the personal identification of person.

By pelvis we can determine the sex.

By long bones we can get to know height estimation.